

Machine Learning Terminology

- **Artificial Intelligence (AI):** the effort to automate intellectual tasks normally performed by humans.
- **Machine Learning (ML):** finding unknown target function: $f: x \rightarrow y$.
- **Deep Learning (DL):** a specific subfield of ML that puts an emphasis on learning successive layers of increasingly meaningful representations.
- **Supervised Learning:** the training set fed to the ML includes the desired solutions, called labels.
- **Unsupervised Learning:** the training data are unlabeled.
- **Semisupervised Learning:** the training data are partially labeled.
- **Self-supervised Learning:** A model is trained on a task using the data itself to generate supervisory signals, rather than relying on external labels provided by humans.
- Categories based on **output**
 - **Classification:** predict a class (e.g., digits 0, 1, ..., 9)
 - **Regression:** predict a value (e.g., house prices)
- **Training dataset:** a collection of data used to train a ML model.
- **Validation dataset:** a subset of the data used to fine-tune a model's hyperparameters and evaluate its performance during training. It acts as an intermediary between training and test datasets to prevent overfitting and ensure the model generalizes well to unseen data.
- **Test dataset:** a separate subset of data used to evaluate the final model's performance on unseen data after training and validation phases.
- **Weights or parameters:** numerical values that the model learns during the training process. They are the fundamental components that determine how the model makes predictions or decisions based on the input data.
- **Loss function:** a function quantifies the difference between the predicted output of a model and the actual target value. It serves as a mathematical guide for how well the model is performing, with the goal being to minimize this loss during training.
- **Optimizer:** an algorithm or a method used to adjust the weights or parameters of a model to minimize the loss function and improve performance.
- **Missing data:** refers to the absence of some values in a dataset where values are expected. It is a common issue that can affect the performance and reliability of machine learning models if not handled properly.
- **Categorical data:** refers to data that represents discrete groups or categories rather than numerical values.
- **Feature scaling:** the process of normalizing or standardizing the range of independent variables (features) in the dataset.
- **Overfitting:** the model performs exceptionally well on the training data but poorly on unseen or test data, failing to generalize to new inputs.
- **Underfitting:** a model is too simple to capture the underlying patterns in the data.